

Student Name: _____

Class Name: **College Algebra FA26 - 014 MW
2:30**Number of Questions: **10**Instructor Name: **Mamani Velasco, Ruben****Question 1 of 10**

A ball is thrown vertically upward. After t seconds, its height h (in feet) is given by the function $h(t) = 72t - 16t^2$. After how long will it reach its maximum height?

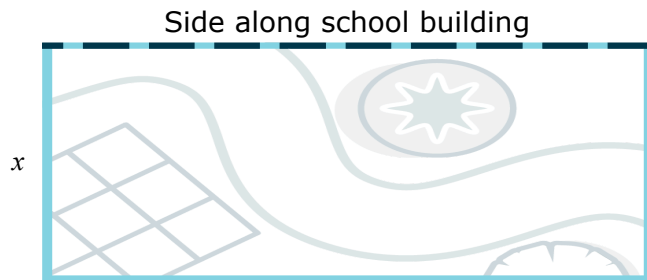
Do not round your answer.

Time: _____ seconds

Question 2 of 10

A school wishes to form three sides of a rectangular playground using 280 meters of fencing. The playground borders the school building, so the fourth side does not need fencing.

As shown below, one of the sides has length x (in meters).



(a) Find a function that gives the area $A(x)$ of the playground (in square meters) in terms of x .

$A(x) =$ _____

(b) What side length x gives the maximum area that the playground can have?

Side length x : _____ meters

(c) What is the maximum area that the playground can have?

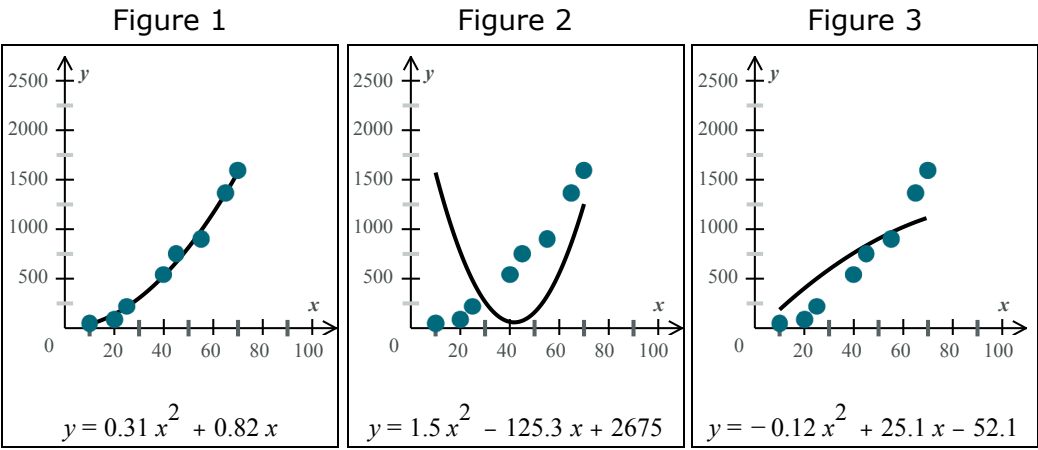
Maximum area: _____ square meters

Question 3 of 10

The data points show the distance y (in meters) required to stop a train traveling at a speed x (in kilometers per hour).

Each figure has the same data points.
However, each figure has a different curve fitting the data.
The equation for each curve is also shown.

Answer the questions that follow.



- (a) Which curve fits the data best?
- ☐ Figure 1 ☐ Figure 2 ☐ Figure 3
- (b) Use the equation of the best fitting curve from part (a) to predict the distance required to stop the train when it is travelling 30 kilometers per hour. Give an exact answer, not a rounded approximation.
- _____ meters

Question 4 of 10

Find all the zeros of the quadratic function.

$$y = x^2 + x - 20$$

If there is more than one zero, separate them with commas.

Question 5 of 10

Suppose that the polynomial function f is defined as follows.

$$f(x) = 6x(x + 7)(x - 8)^2(x + 12)^3$$

List each zero of f according to its multiplicity in the categories below.

Question 6 of 10

Find all y -intercepts and x -intercepts of the graph of the function.

$$f(x) = x^3 + 3x^2 - 4x - 12$$

If there is more than one answer, separate them with commas.

Write "None" if applicable.

y -intercept(s):

x -intercept(s):

Question 7 of 10

Choose the end behavior of the graph of each polynomial function.

(a) Choose the end behavior for $f(x) = 6x^5 - 9x^3 + 4x^2 + x$.

- ☐ The graph of f falls to the left and rises to the right.
- ☐ The graph of f rises to the left and falls to the right.
- ☐ The graph of f rises to the left and rises to the right.
- ☐ The graph of f falls to the left and falls to the right.

(b) Choose the end behavior for $f(x) = -6x^3 - 3x^2 + 5x - 1$.

- ☐ The graph of f falls to the left and rises to the right.
- ☐ The graph of f rises to the left and falls to the right.
- ☐ The graph of f rises to the left and rises to the right.
- ☐ The graph of f falls to the left and falls to the right.

(c) Choose the end behavior for $f(x) = -4x(x - 4)(x + 2)$.

- ☐ The graph of f falls to the left and rises to the right.
- ☐ The graph of f rises to the left and falls to the right.
- ☐ The graph of f rises to the left and rises to the right.
- ☐ The graph of f falls to the left and falls to the right.

Question 8 of 10

Consider the following polynomial function.

$$f(x) = (x + 1)(x - 1)(x - 3)$$

Answer the questions regarding the graph of f .
Then, use this information to graph the function.

(a) Choose the end behavior of the graph of f .

(Blank 1)

Blank 1 Options

- Falls to the left and rises to the right
- Rises to the left and falls to the right
- Rises to the left and rises to the right
- Falls to the left and falls to the right

(b) List each real zero of f according to the behavior of the graph at the x -axis near that zero. If there is more than one answer, separate them with commas. If there is no answer, write "None".

Zero(s) where the graph crosses the x -axis:

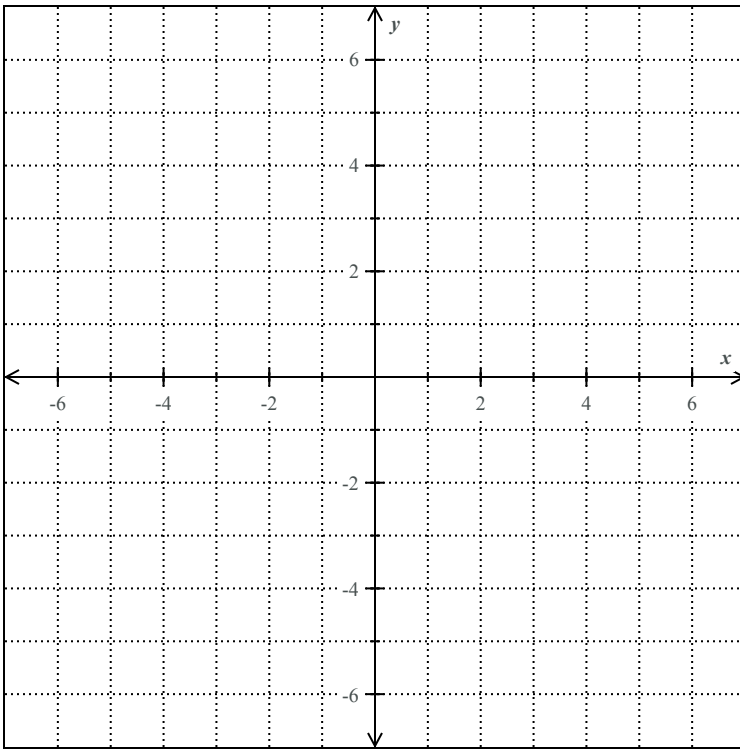
Zero(s) where the graph touches, but does not cross the x -axis:

(c) Find the y -intercept of the graph of f .

y -intercept:

(d) Graph $f(x) = (x + 1)(x - 1)(x - 3)$ by doing the following.

- Plot all points where the graph of f intersects the x -axis or y -axis.
- For each point on the x -axis, select the correct behavior.
- Connect the points following the behaviors.



Question 9 of 10

Consider the following polynomial function.

$$f(x) = -(x - 1)(x + 2)(x + 1)^2$$

Answer the questions regarding the graph of f .
Then, use this information to graph the function.

(a) Choose the end behavior of the graph of f .

(Blank 1)

Blank 1 Options

- Falls to the left and rises to the right
- Rises to the left and falls to the right
- Rises to the left and rises to the right
- Falls to the left and falls to the right

(b) List each real zero of f according to the behavior of the graph at the x -axis near that zero. If there is more than one answer, separate them with commas. If there is no answer, write "None".

Zero(s) where the graph crosses the x -axis:

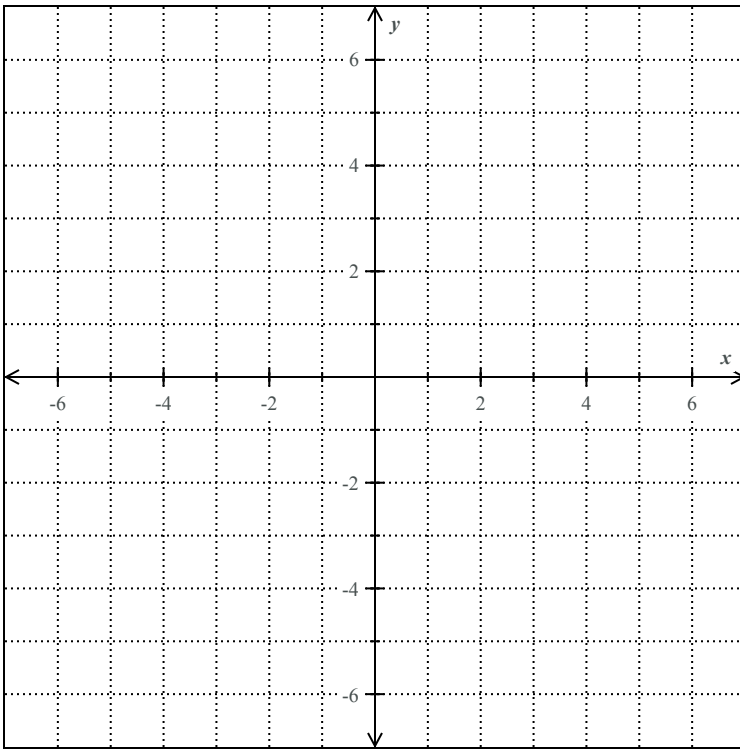
Zero(s) where the graph touches, but does not cross the x -axis:

(c) Find the y -intercept of the graph of f .

y -intercept:

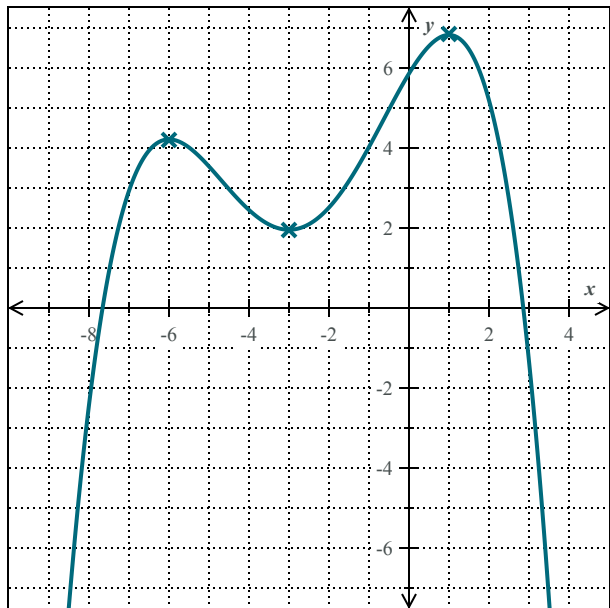
(d) Graph $f(x) = -(x - 1)(x + 2)(x + 1)^2$ by doing the following.

- Plot all points where the graph of f intersects the x -axis or y -axis.
- For each point on the x -axis, select the correct behavior.
- Connect the points following the behaviors.



Question 10 of 10

Below is the graph of a polynomial function with real coefficients. All local extrema of the function are shown in the graph.



Use the graph to answer the following questions.

(a) Over which intervals is the function increasing? Choose all that apply.

☐ $(-\infty, -6)$ ☐ $(-6, -3)$ ☐ $(-3, 1)$ ☐ $(-6, 1)$ ☐ $(1, \infty)$

(b) At which x -values does the function have local maxima? If there is more than one value, separate them with commas.

(c) What is the sign of the function's leading coefficient?

Positive Negative Not enough information

(d) Which of the following is a possibility for the degree of the function? Choose all that apply.

☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ 8 ☐ 9

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Number of Questions: **10**

Question 1 of 10

Time: 2.25 seconds

Question 2 of 10

(a) $A(x) = 280x - 2x^2$

(b) Side length x : 70 meters

(c) Maximum area: 9800 square meters

Question 3 of 10

(a) Which curve fits the data best?

☒ Figure 1 ☐ Figure 2 ☐ Figure 3

(b) Use the equation of the best fitting curve from part (a) to predict the distance required to stop the train when it is travelling 30 kilometers per hour. Give an exact answer, not a rounded approximation.

303.6 meters

Question 4 of 10

zero(s): 4, -5

Question 5 of 10

Zero(s) of multiplicity one: 0, -7

Zero(s) of multiplicity two: 8

Zero(s) of multiplicity three: -12

Question 6 of 10

y -intercept(s): -12

x -intercept(s): $-3, 2, -2$

Question 7 of 10

- (a) ☒ The graph of f falls to the left and rises to the right.
☐ The graph of f rises to the left and falls to the right.
☐ The graph of f rises to the left and rises to the right.
☐ The graph of f falls to the left and falls to the right.
- (b) ☐ The graph of f falls to the left and rises to the right.
☒ The graph of f rises to the left and falls to the right.
☐ The graph of f rises to the left and rises to the right.
☐ The graph of f falls to the left and falls to the right.
- (c) ☐ The graph of f falls to the left and rises to the right.
☒ The graph of f rises to the left and falls to the right.
☐ The graph of f rises to the left and rises to the right.
☐ The graph of f falls to the left and falls to the right.

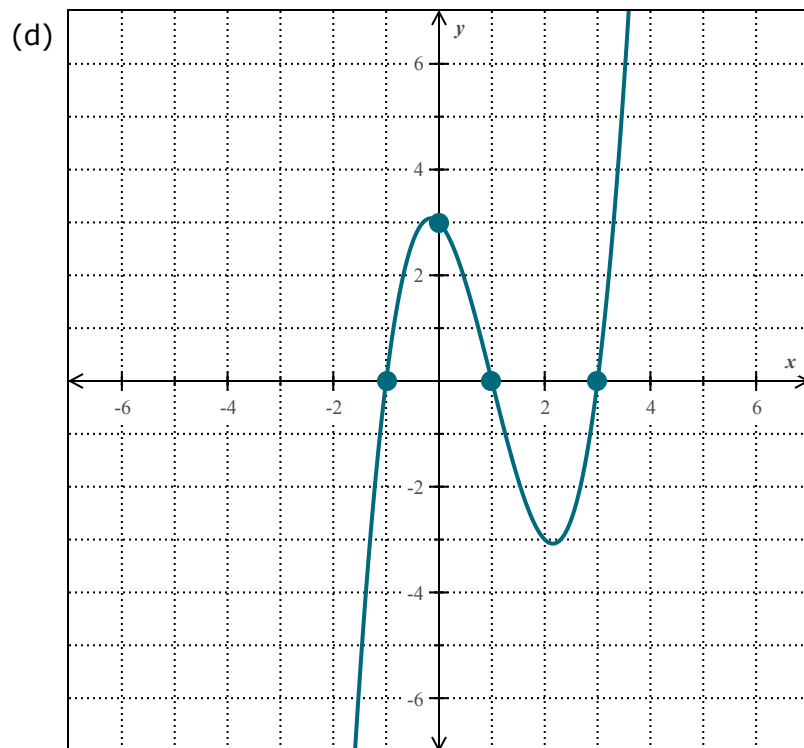
Question 8 of 10

(a) Falls to the left and rises to the right ▼

(b) Zero(s) where the graph crosses the x -axis: $-1, 1, 3$

Zero(s) where the graph touches, but does not cross the x -axis: No Solution

(c) y -intercept: 3



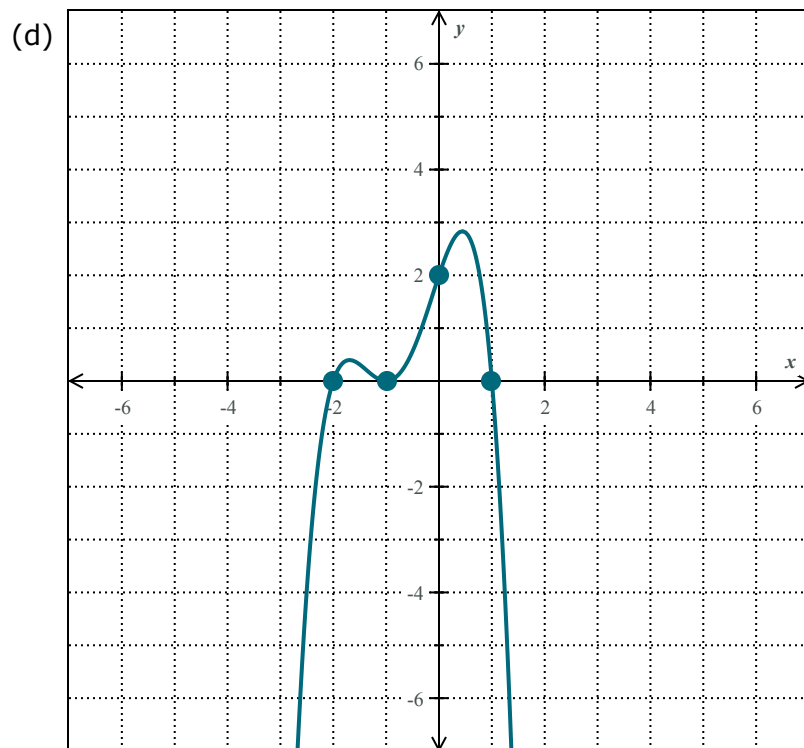
Question 9 of 10

(a) Falls to the left and falls to the right ▼

(b) Zero(s) where the graph crosses the x -axis: $1, -2$

Zero(s) where the graph touches, but does not cross the x -axis: -1

(c) y -intercept: 2



Question 10 of 10

(a) Over which intervals is the function increasing? Choose all that apply.

☒ $(-\infty, -6)$ ☐ $(-6, -3)$ ☒ $(-3, 1)$ ☐ $(-6, 1)$ ☐ $(1, \infty)$

(b) At which x -values does the function have local maxima? If there is more than one value, separate them with commas.

$-6, 1$

(c) What is the sign of the function's leading coefficient?

Negative

(d) Which of the following is a possibility for the degree of the function? Choose all that apply.

☒ 4 ☐ 5 ☒ 6 ☐ 7 ☒ 8 ☐ 9